Model-Based Decision Making
MSc Complex Systems & Policy

Assignment 1:
Telegram Mention Network

Pro-Kremlin and Anti-Kremlin Channels

Study period: 24 February – 24 May 2022

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1 Introduction and Dataset

This analysis examines the Telegram mention network between "Pro-Kremlin" and "Anti-Kremlin" channels during the early phase of Russia's full-scale invasion of Ukraine. The dataset was compiled by Bawa et al. [2]. The original paper scraped posts across 404 Pro-Kremlin and 114 Anti-Kremlin channels, manually classifying them by their stance on state policy and the war.

For this analysis, the time frame was set from 24 February to 24 May 2022 to study network structure during the rapid increase in posting volume and channel activity [2]. After removing four duplicate usernames, 400 unique "Pro-Kremlin" and 114 "Anti-Kremlin" channels remained (in total 514 nodes), including 75 isolated nodes. Each directed edge represents a channel mention, weighted by the number of posts mentioning the target (counted once per post). Self-mentions and mentions of channels outside the selected lists were excluded. The directed, weighted graph (G) contains 8,581 edges. Its undirected, unweighted representation (U) contains 7,449 edges and is used for topological analysis, model benchmarking and robustness analysis.

2 Empirical Topological Analysis

The network is sparse (density = 0.0565), meaning most pairs of channels are not directly connected. The average degree is 28.98. However, because the network includes 75 isolated nodes (14.6%), this average may not fully reflect the connectivity of a typical node (Table 1). The network has 76 connected components, but it is dominated by a single large component (LCC) containing 85.4% of all channels (439/514). Inside this main component, the average path length is short (2.37 steps) and the diameter is small (Table 2). This shows that the core of this network is mostly compact, with short topological distances between channels. However, these numbers measure structural paths rather than actual message forwarding.

The network shows strong local clustering, with a global transitivity of 0.377 and a mean local clustering coefficient of 0.352. This structure indicates that channels tend to form tight clusters. In contract, degree assortativity is negative ($r_{\text{degree}} = -0.098$), which reveals that high-degree channels slightly tend to connect with low-degree ones. Meanwhile, the channels tend to connect within their own ideological group ($r_{\text{group}} = 0.376$). Network reciprocity is 0.264, indicating that 26.4% of directed channel ties are mutual (Table 2).

The distribution of weighted mentions reveals a strong group asymmetry. Interactions within the Pro-Kremlin group account for 93.1% of all recorded mentions. However, Anti-Kremlin channels direct nearly a third of their references (32.9%) to Pro-Kremlin targets. In contrast, only 2.5% of Pro-Kremlin mentions target Anti-Kremlin channels, even though this small share translates to a higher absolute volume (1,215 vs. 750 mentions). Overall, mentions reflect visibility rather than ideological endorsement.

As shown in Figures 1 and 2, the network's degree distribution is strongly right-skewed**.** Most connected channels have between 1 and 30 connections, whereas a small number reach up to 183. The log–log CCDF displays a downward bend instead of a linear decrease, which argues against a power-law model, showing that the network might not exhibit scale-free properties.

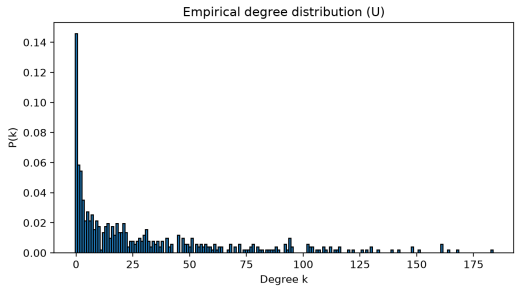


Figure 1: Histogram $P(k)$.

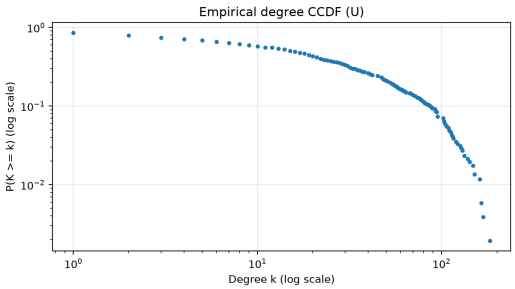


Figure 2: CCDF on log–log axes.

Table 1: Basic Network Metrics

Metric	Value
Nodes (N)	514
Edges (M)	7,449
Density	0.0565
Average degree	28.98
Isolated nodes	75 (14.6%)
Connected components	76
LCC	439 nodes (85.4%)
Average shortest path (LCC)	2.37

Table 2: Advanced Topological Properties

Metric	Value
Diameter (LCC)	6
Maximum degree	183
Global transitivity	0.377
Average local clustering	0.352
Degree assortativity (r_{degree})	−0.098
Attribute assortativity (r_{group})	0.376
Reciprocity (directed graph G)	0.264

3 Model Network Benchmarking

To evaluate these topological properties, the network was compared against standard theoretical models [3, 4, 1] of matched size ($N = 514$) and average degree. Parameters were set to $p = 0.0565$ for Erdős–Rényi (ER), $k = 28$ and $p = 0.1$ for Watts–Strogatz (WS), and $m = 14$ for Barabási–Albert (BA). While the ER model closely mirrors empirical density, the WS and BA variants generate slightly fewer edges due to degree-parity and integer constraints. The comparison results are summarised in Table 3.

Clustering provides the clearest contrast between the models. The empirical average clustering coefficient (0.3515) is approximately 6.3 times higher than the ER value (0.0561). The WS model produces substantially higher clustering (0.5346), whereas the BA model produces lower clustering (0.1199). Consequently, none of the three models reproduces the empirical clustering level.

Path lengths show a different pattern. The empirical average shortest path length (2.3717) is relatively close to ER (2.1306) and BA (2.1650), while WS produces longer paths (2.6070). Regarding degree assortativity, the empirical network is more disassortative (r = -0.0981) than all three benchmarks. BA is generally closer to the empirical value (r = -0.0305), but none of the models fully represent this tendency.

Figure 3 shows the differences in **degree distribution** across models. The three benchmark models drop sharply in a narrow range, reflecting homogeneous or bounded degrees. In contrast, the empirical network stretches up to $k = 183$, demonstrating a much higher degree heterogeneity and structural resilience than standard random or scale-free approximations can capture.

Table 3: Empirical network and benchmark model comparison.

Model	N	M	$\langle k \rangle$	$\langle C \rangle$	$\langle L \rangle$ (LCC)	r	LCC share
Empirical (U)	514	7,449	28.9844	0.3515	2.3717	−0.0981	0.8541
ER (mean ± std, 20 runs)	514	7,403.6 ± 51.6	28.8078 ± 0.2006	0.0561 ± 0.0009	2.1306 ± 0.0047	−0.0020 ± 0.0098	1.0000 ± 0.0000
WS (mean ± std, 20 runs)	514	7,196.0 ± 0.0	28.0000 ± 0.0000	0.5346 ± 0.0043	2.6070 ± 0.0068	−0.0054 ± 0.0109	1.0000 ± 0.0000
BA (mean ± std, 20 runs)	514	7,000.0 ± 0.0	27.2374 ± 0.0000	0.1199 ± 0.0035	2.1650 ± 0.0047	−0.0305 ± 0.0089	1.0000 ± 0.0000

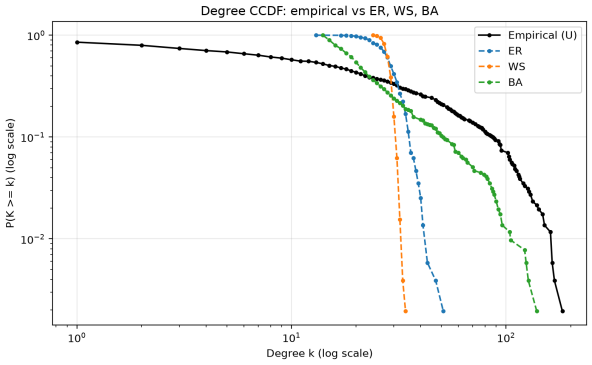


Figure 3: Degree CCDF comparison between empirical network and benchmark models.

4 Centrality and Node Importance

To identify structurally important channels, four centrality measures were calculated on the directed graph (G): degree centrality (direct connections), betweenness (position on shortest paths), closeness-in (how easily a channel can be reached from others) and weighted PageRank (which also accounts for mention frequency). Table 4 presents the top ten channels for each measure.

The top ten channels by degree centrality, betweenness and closeness-in consist entirely of Pro-Kremlin channels. This indicates that these channels have many direct connections and occupy central positions in the network's structure. PageRank also ranks mostly Pro-Kremlin channels but includes two Anti-Kremlin channels: @*teamnavalny* and @*navalny*. Overall, 80% of the PageRank top ten are Pro-Kremlin, close to their share of the entire network (77.8%). Notably, closeness-in measures how easily a channel can be reached through incoming directed paths, rather than its ability to spread information outward.

The measures are positively correlated (Spearman's $\rho = 0.671$–0.861), with the strongest associations between degree centrality and closeness-in (0.861), and closeness-in and PageRank (0.858). Degree centrality and betweenness are also strongly correlated (0.798), sharing nine channels in their top-ten rankings. In contrast, PageRank and betweenness share only one channel, suggesting that being frequently mentioned by important channels does not necessarily mean occupying a key position on shortest paths.

Several channels rank in the top twenty by betweenness but not by total degree, including @*kaktovottak*, @*vchkogpu* and several others. Their rankings suggest potential structural brokerage, as they occupy important positions on shortest paths despite ranking lower by degree. However, this does not necessarily mean they connect different ideological groups or control information flows.

Table 4: Top 10 channels by degree centrality, betweenness, closeness-in, and weighted PageRank.

Rank	Degree centrality	Betweenness	Closeness-in	Weighted PageRank
1	rus_demiurge (0.4542)	russica2 (0.0405)	rybar (0.3628)	rvvoenkor (0.0543)
2	dr_alex_sosnowski (0.4289)	sashakots (0.0320)	sashakots (0.3590)	rusvesnasu (0.0387)
3	sashakots (0.4269)	solovievlive (0.0313)	opersvodki (0.3424)	sashakots (0.0367)
4	solovievlive (0.4269)	karaulny (0.0273)	bbbreaking (0.3404)	rybar (0.0364)
5	sergeykolyasnikov (0.4055)	kaktovottak (0.0272)	boris_rozhin (0.3341)	opersvodki (0.0276)
6	t34newsfeed (0.3879)	rus_demiurge (0.0270)	readovkanews (0.3336)	mdaudov_95 (0.0156)
7	boris_rozhin (0.3859)	t34newsfeed (0.0255)	dimsmirnov175 (0.3325)	glavpolit (0.0145)
8	karaulny (0.3626)	sergeykolyasnikov (0.0254)	voenkorkotenok (0.3294)	teamnavalny (Anti) (0.0141)
9	russica2 (0.3626)	boris_rozhin (0.0233)	mig41 (0.3283)	mig41 (0.0138)
10	rybar (0.3392)	dr_alex_sosnowski (0.0199)	solovievlive (0.3262)	navalny (Anti) (0.0132)

5 System Dynamics and Policy Implications

Regarding network **robustness**, the largest connected component (LCC) was tested under random node removal (averaged over 20 runs) and targeted removal based on initial degree, initial betweenness and adaptive degree.

As illustrated in Figure 4, the LCC fell below 50% of the original network size after 40% of random removals, compared with 28% for degree, 24% for betweenness and 26% for adaptive removal. This shows that the network is more vulnerable to targeted attacks than to random failures. However, the initial LCC already contains only 85.4% of all nodes, and these results reflect structural connectivity rather than actual network functioning.

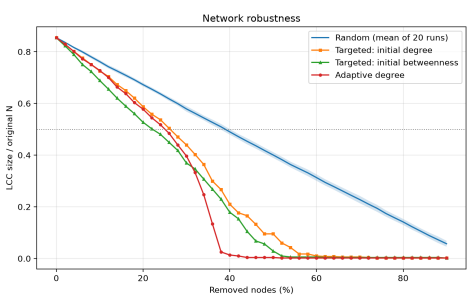


Figure 4: Network robustness analysis.

The network also shows **small-world properties**, combining high clustering (C = 0.3515), approximately 6.3 times higher than ER (0.0561), with relatively short average path lengths (L = 2.37 compared with 2.13 for ER). Its diameter is only 6. These findings suggest a small-world-like structure within

the main component. However, the path lengths are not directly comparable because the empirical LCC contains 439 nodes, while ER contains 514.

Several **generative mechanisms** may explain this structure. Positive group assortativity (r_group = 0.376) and within-group mentions suggest ideological homophily. High transitivity may be related to triadic closure, while the right-skewed degree distribution may reflect cumulative advantage or preferential attachment. However, these mechanisms were not directly tested.

Regarding **information diffusion**, the short paths within the LCC suggest that channels are structurally close to one another. Positive group assortativity and within-group mentions indicate that attention is largely concentrated within ideological groups. At the same time, cross-group mentions, particularly from Anti- to Pro-Kremlin channels, suggest potential connections between these groups. Channels with relatively high betweenness may also act as structural bridges. However, mentions do not necessarily represent forwarding or actual information transmission.

Finally, the **policy implications** include identifying highly connected hubs and potential bridges, examining cross-group connections, and considering the content of interactions when assessing channel influence. For example, @*kaktovottak* and @*vchkogpu* have relatively high betweenness despite ranking lower in total degree. These channels may warrant further investigation as potential structural bridges. However, centrality alone does not establish political influence or endorsement.

6 Limitations

The analysis has several limitations. First of all, the dataset includes only channels from the selected Pro-Kremlin and Anti-Kremlin classification lists, which exclude all other channels out the scope. Moreover, the groups are unequal (400 Pro-Kremlin and 114 Anti-Kremlin channels), which limits direct comparisons. Similarly, shares of mentions within each source group do not fully account for differences in group size or posting activity. Furthermore, mentions may indicate endorsement, criticism or neutral reference, which cannot be distinguished using this dataset. Edge direction and weight therefore describe observed attention rather than agreement.

Data cleaning also involves limitations. After removing rows with missing channel or message information and filtering by date 963,574 rows remained. Of these, 15,352 came from channels outside the selected lists and 1,209 came from files belonging to the wrong group. Both were excluded. Additionally, forwarded messages were not included, meaning that some instances of content being shared between channels may be missing from the network.

The study covers a limited period (24 February–24 May 2022) and does not capture changes in network structure over longer periods or other phases of the conflict. Clustering, assortativity and path-based metrics are calculated on the undirected, unweighted graph, which ignores edge direction and mention frequency. Path lengths are calculated on the empirical LCC (439 nodes), whereas the benchmark models are connected and include all 514 nodes, which limits direct comparability. Betweenness centrality is calculated exactly, but all centrality measures describe structural positions rather than directly measured political influence. Finally, the ER, WS and BA models represent only three simplified generative mechanisms; other models incorporating group structure or homophily may capture additional empirical properties.

7 Conclusion

The mention network of selected Pro-Kremlin and Anti-Kremlin Telegram channels during February–May 2022 is sparse but highly clustered, with a large connected component containing 85.4% of all channels. Its structure exhibits positive group assortativity and a small number of highly central channels, most of which are Pro-Kremlin. Cross-group attention is asymmetric when normalised by source-group activity: 32.9% of Anti-Kremlin mentions target Pro-Kremlin channels, compared with 2.5% in the reverse direction, although the absolute number of Pro-to-Anti mentions is higher.

None of the ER, WS or BA models reproduces the combination of structural characteristics observed in the empirical network. ER most closely approximates its average degree and path length,

while WS produces excessive clustering and BA captures some degree heterogeneity but not the full empirical structure. Robustness simulations also show greater vulnerability to targeted node removal than to random failures. Overall, the findings describe patterns of attention and structural connectivity rather than political influence or actual information diffusion. Further research incorporating message content, forwarding data and temporal network dynamics would provide a more complete understanding of interactions between these channels.

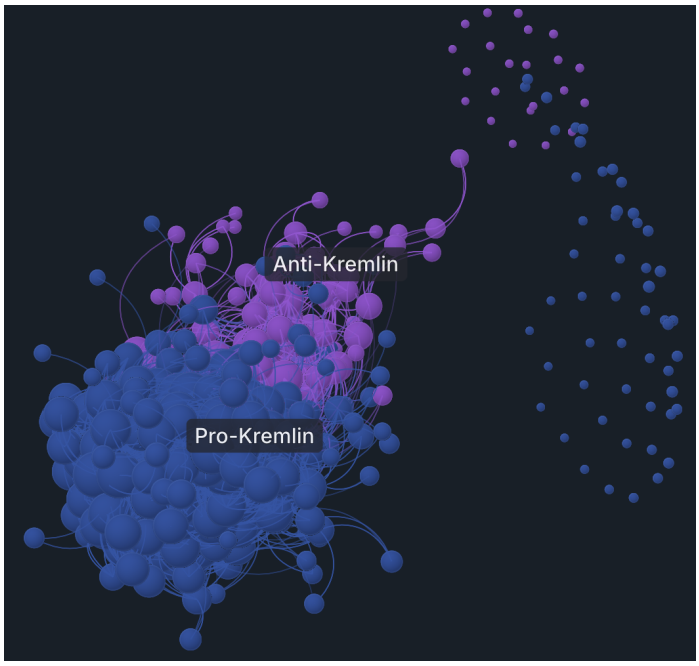


Figure 5: 3D visualisation of the Telegram mention network created using cosmograph.app.

References

[1] Barabási, A.-L. (2016). *Network science*. Cambridge University Press.

[2] Bawa, A., Kursuncu, U., Achilov, D., Shalin, V. L., Agarwal, N., & Akbas, E. (2025). Telegram as a battlefield: Kremlin-related communications during the Russia–Ukraine conflict. *Proceedings of the International AAAI Conference on Web and Social Media, 19*(1), 2361–2370. `https://doi.org/10.1609/icwsm.v19i1.35939`

[3] Newman, M. E. J. (2018). *Networks: An introduction* (2nd ed.). Oxford University Press.

[4] Watts, D. J., & Strogatz, S. H. (1998). Collective dynamics of small-world networks. *Nature, 393*, 440–442.

Data and Code Availability

The analysis is based on the Telegram dataset provided by Bawa et al. (2025), available at Figshare. The code and analysis are available on GitHub. An interactive 3D visualisation of the network is available on Cosmograph.

AI Statement

AI tools were used to assist with LaTeX formatting and code review. All analytical interpretations were written by the author.